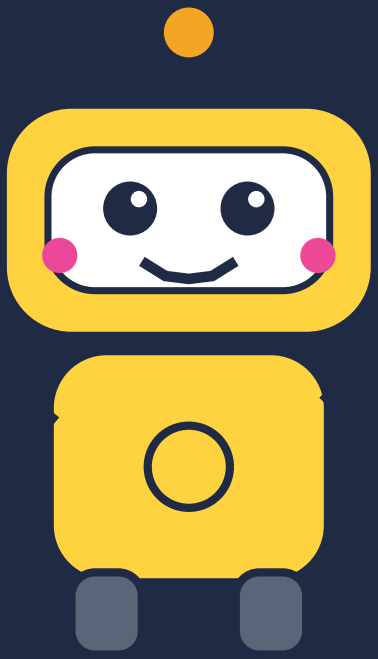
TEACHER BOOK

Think Like a Coder

Teacher book for Class 4

A 5-period plan for every chapter
Hands-on activities, common mistakes and exit checks
Answer keys, assessment guide and photocopiable journal



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 4 · Follows the 14 chapters of the NCERT Class 4 maths textbook

SAMPLE EDITION · DRAFT

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About this teacher book

This book goes with the *Think Like a Coder* Class 4 workbook. It gives you a plan for every chapter, so that a maths or class teacher can teach computational thinking confidently without being a computer specialist. CBSE suggests about **50 hours** a year for Class 4; the plans here use **75 periods of 40 minutes**, which is about 50 hours.

★ What is in this book

- ✓ **A year plan** showing which chapter to teach in which term, next to the NCERT maths chapter.
- ✓ **A 5-period plan for every chapter**: what you do and what students do in each period.
- ✓ **A hands-on activity** for every chapter, with set-up steps and a tip.
- ✓ **Common mistakes** to watch for, and how to support or stretch students.
- ✓ **An exit check** with answers, and what to note in the observation journal.
- ✓ **The answer key** for every page of the student workbook.
- ✓ **Photocopiable** observation journal and term summary pages at the back.

✓ Before you start

- ✓ Read the chapter in the student book and these notes. Most lessons need 5 to 10 minutes to prepare.
- ✓ Get the kit items ready (listed under *At a glance*). Everything is reusable.
- ✓ Plan pairs in advance: mix confident readers with children who prefer talking and acting out.
- ✓ Teach the CT chapter in the same week or fortnight as the matching NCERT maths chapter, so the maths is fresh.
- ✓ Keep it active: children should talk, act out and build more than they listen.

No computer lab needed

No computers are needed. Every idea is taught with paper, cards, the floor mat and talk. These are the same ideas children will later use in block coding and AI: steps in order, loops, decisions, data and testing. A projector helps for showing pages but is optional.

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How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.

2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.

3. **Modify** 8 min
   Children change the steps to do something new.

4. **Put in order** 8 min
   Jumbled code cards go back in the right order.

5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

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Year plan

14 chapters of 5 periods each, plus 5 periods for revision and assessment: **75 periods** of 40 minutes in all. Teach each chapter alongside the NCERT maths chapter with the same name; the order and timing can follow your school's maths plan.

Ch	Chapter	Coder skill	Level	Periods	Term
1	Shapes Around Us	Faces, edges and nets	B1 review	5	Term 1
2	Hide and Seek	If-then-else choices	B3	5	Term 1
3	Patterns Around Us	Odd, even and checking for mistakes	B3	5	Term 1
4	Thousands Around Us	Big numbers and sorting machines	B4	5	Term 1
5	Sharing and Measuring	Fractions and checking a condition	B3	5	Term 1
6	Measuring Length	Shortest paths and loops with turns	B4	5	Term 1
7	The Cleanest Village	Tracing a machine step by step	B3	5	Term 1
8	Weigh It, Pour It	If-else decisions and measuring steps	B3	5	Term 2
9	Equal Groups	Loops inside loops	B4	5	Term 2
10	Elephants, Tigers and Leopards	Clue tables and data	B3	5	Term 2
11	Fun with Symmetry	Mirror patterns and repeat with turns	B4	5	Term 2
12	Ticking Clocks and Turning Calendar	Repeat until and cycles	B4	5	Term 2
13	The Transport Museum	Decision trees: sorting by questions	B3	5	Term 2
14	Data Handling	Collecting and showing data	B2 review	5	Term 2
	Revision and term assessment	Review games, the term summary and parent feedback	All	5	Both

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Assessment and the observation journal

CBSE asks for continuous, competency-based assessment in CT&AI: written puzzles, group activities and a **Teacher Observation Journal**. This book makes that simple. In each chapter, the exit check and the 'what to note' box tell you what to look for; record one level per child per skill every few chapters, not every lesson.

Skill	Beginning	Developing	Secure	Extending
Following steps	Follows short steps with help	Traces steps and machines with prompts	Traces loops, nested loops and states alone	Designs and debugs own programs
Patterns	Spots simple repeats	Continues patterns and spots odd and even	Uses patterns to check for mistakes	Makes new patterns and explains them
Breaking problems down	Needs help to start	Breaks familiar jobs into steps	Plans steps such as pouring or routes	Finds the shortest or best plan
Logic and rules	Uses one clue	Follows an if-else rule	Writes if-else rules and uses clue tables	Builds decision trees and explains choices

How to assess without extra tests

- **Watch and listen** during activities: explaining, predicting and fixing show understanding better than a written answer.
- **Use the exit check** at the end of each chapter: two quick questions, answers in this book.
- **Compare with self-checks**: each chapter ends with 'How did I do?', where children colour a face for each 'I can' statement.
- **Summarise each term** on the term summary page: one level per area and a short comment for the report card.

The four levels

Use the four levels for every skill. **Beginning**: needs help to start. **Developing**: succeeds with prompts or a partner. **Secure**: succeeds alone on familiar problems. **Extending**: solves new problems and explains why. Most children will be Developing or Secure; that is expected and healthy.

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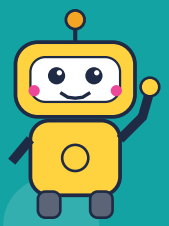
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TEACHER NOTES · CHAPTER 1 · LEVEL B1 review

Shapes Around Us

Coder skill: Faces, edges and nets

Goes with NCERT maths, Class 4, chapter 1



At a glance	
Students will	count faces, edges and corners, find opposite faces in a net and follow drawing steps exactly.
Key words	**face** (a flat side of a solid), **edge** (where two faces meet), **net** (a flat shape that folds into a solid)
Materials	Empty boxes (cuboids), a dice or cube, squared paper, scissors, glue
Time	5 periods of 40 minutes
Link	Teach alongside chapter 1 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *face*, *edge*, *net*. Work through the example *Which faces are opposite?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Open the box	Run the activity below. Move between groups and note who can explain the idea.	Count faces, edges and corners, flatten a box into a net, then draw, fold and check a cube net.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write drawing steps so a friend can draw a cube net on squared paper. Model the first card or step, then let pairs finish.	Follow the steps, cut out the net and fold it. Does it make a cube? Write which faces are opposite.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Open the box

1. Give each group an empty box. Count its faces, edges and corners together, and record them.
2. Cut the box along some edges and flatten it. Ask: 'Which flat shape did we get?' That is a net.
3. On squared paper, groups draw the cross-shaped cube net, cut it out and fold it.
4. Before folding, they colour the pairs of faces they think will be opposite. Fold and check.

Tip: Let children predict opposite faces before folding; checking a prediction is the habit we want.

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Common mistakes to watch for

- **Counting the same edge twice**, or missing the hidden corners at the back. Mark each one as you count.
- **Thinking any 6 squares make a cube net.** A 2 by 3 block does not fold into a cube.
- **Steps that are not exact**, such as 'draw some squares'. Say how many and where.

Different paces

Support: Use a real dice and count by touching each face.
Stretch: Find a different net that also folds into a cube, and one that does not; explain the difference.

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many faces, edges and corners does a cube have? (Answer: 6 faces, 12 edges, 8 corners)

2. Does a 2 by 3 block of 6 squares fold into a cube? (Answer: no)

Note in the journal: Beginning (needs help to count faces, edges and corners); **Developing** (can count faces, edges and corners with prompts or a partner); **Secure** (can count faces, edges and corners and find opposite faces in a net alone); **Extending** (can follow drawing steps exactly and explain why).

Answer key

1. 6 faces, 12 edges, 8 corners
2. c) D
3. b) the 2 by 3 block
4. 5
5. a
Build it: Yes: the cross net folds into a cube; faces 1 and 3, 2 and 4 in the column, and the two side squares, are opposite
Challenge: 1 (the cube in the very middle)

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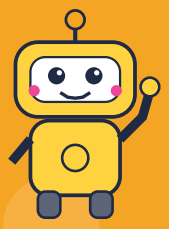
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TEACHER NOTES · CHAPTER 2 · LEVEL B3

Hide and Seek

Coder skill: If-then-else choices

Goes with NCERT maths, Class 4, chapter 2



At a glance	
Students will	follow an if-then-else rule, trace a program with a choice and write my own if-else program.
Key words	**if-then-else** (a choice between two actions), **condition** (the question that is true or false)
Materials	Floor tiles or a strip of 5 squares (some marked sun, some snow), Robo card, if-else code cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 2 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *if-then-else*, *condition*. Work through the example *Robo walks along the path* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Robo makes choices	Run the activity below. Move between groups and note who can explain the idea.	Act out an if-else inside a loop, predict what Robo says on each square, and write their own rule.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a program that makes Robo say 'Hot!' on sun tiles and 'Cold!' on snow tiles along a path of 5 tiles. Model the first card or step, then let pairs finish.	Draw a path of 5 sun and snow tiles. Swap with a friend and trace each other's program.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Robo makes choices

1. Lay out a strip of 5 squares, some marked sun and some snow. A child is Robo.
2. Hold up the program: *repeat 5 times: move forward; if on a sun square then say 'Hot!' else say 'Cold!'*
3. Robo walks slowly. On every square the class asks the condition aloud: 'Is it a sun square?'
4. Change the strip and predict what Robo will say before walking it again.

Tip: Ask 'Did Robo say Hot AND Cold on the same square?' It shows that only one branch runs each time.

Common mistakes to watch for

- **Thinking both branches run.** Only one of 'then' and 'else' happens each time.
- **Thinking the 'if' is checked only once.** Inside a loop it is checked on every square.
- **Forgetting the else.** Ask what Robo says on a snow square if there is no else.

Different paces

Support: Use 3 squares and say the condition aloud on each one.
Stretch: Add a third kind of square (rain) and handle three choices with if, else if, else.

Exit check and what to note

Two quick questions at the end of the chapter:

1. On the path sun, snow, snow, what does Robo say? (Answer: Hot!, Cold!, Cold!)

2. In an if-else, can both the 'then' part and the 'else' part happen at the same time? (Answer: no, only one of them)

Note in the journal: Beginning (needs help to follow an if-then-else rule); **Developing** (can follow an if-then-else rule with prompts or a partner); **Secure** (can follow an if-then-else rule and trace a program with a choice alone); **Extending** (can write my own if-else program and explain why).

Answer key

1. Step, Splash, Splash, Step
2. jump, clap, jump
3. c) 4
4. b) blue
5. a) C2
Build it: Repeat 5 times: move forward; if on a sun tile then say 'Hot!' else say 'Cold!'
Challenge: 3 (tiles 3, 6 and 9)

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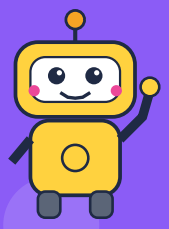
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TEACHER NOTES · CHAPTER 3 · LEVEL B3

Patterns Around Us

Coder skill: Odd, even and checking for mistakes

Goes with NCERT maths, Class 4, chapter 3



At a glance	
Students will	tell odd from even, add a check coin and find a flipped coin.
Key words	**even** (shares into pairs), **odd** (one left over), **parity check** (an even/odd check for mistakes)
Materials	16 two-coloured counters or coins per group (or paper circles coloured on one side)
Time	5 periods of 40 minutes
Link	Teach alongside chapter 3 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *even*, *odd*, *parity check*. Work through the example *Coin Flip Magic* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Coin flip magic	Run the activity below. Move between groups and note who can explain the idea.	Perform and explain a parity-check card trick, and find a flipped coin using odd and even counts.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Put the Coin Flip Magic steps in order, then perform the trick for a friend. Model the first card or step, then let pairs finish.	Number the cards 1 to 4, then try the trick at home with buttons or paper circles.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Coin flip magic

1. One child lays out a 4 by 4 grid of coins, heads or tails up in any order.
2. The 'magician' quietly adds a fifth coin to each row and column so that every row and column has an even number of heads.
3. While the magician looks away, a friend flips one coin.
4. The magician finds the row and the column that are now odd: the flipped coin is where they cross. Then explain the trick to the class.

Tip: Practise the trick yourself before class. Reveal the secret only after children have guessed: they enjoy working it out.

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Common mistakes to watch for

- **Making rows even but forgetting the columns.** Both need check coins.
- **Counting tails instead of heads** halfway through. Agree what you count before starting.
- **Thinking odd means 'one'.** 3 and 5 are odd too; odd means one left over after pairing.

Different paces

Support: Use a 3 by 3 grid and count in pairs.
Stretch: What happens if two coins are flipped? Can the magician still find them? (Not always: that is why bigger checks exist.)

Exit check and what to note

Two quick questions at the end of the chapter:

1. A row has 3 heads. Is it odd or even, and what should the check coin be? (Answer: odd; the check coin must be heads, to make 4)

2. After one coin is flipped, row 2 and column 3 are odd. Where is the flipped coin? (Answer: row 2, column 3)

Note in the journal: Beginning (needs help to tell odd from even); **Developing** (can tell odd from even with prompts or a partner); **Secure** (can tell odd from even and add a check coin alone); **Extending** (can find a flipped coin and explain why).

Answer key

1. 27 and 41
2. Row 1 white, row 2 black, row 3 black
3. b) row 2, column 1 (row 2 and column 1 are now odd)
4. a) even
5. 32 (double each time)
Build it: Lay out a 4 x 4 grid; add check coins to make each row and column even; friend flips one coin; find the odd row and column
Challenge: No: the row stays even and two columns turn odd, so you know something changed but not exactly which coins

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4

TEACHER NOTES · CHAPTER 4 · LEVEL B4

Thousands Around Us

Coder skill: Big numbers and sorting machines

Goes with NCERT maths, Class 4, chapter 4



At a glance	
Students will	read numbers in the thousands, order big numbers and use a sorting network.
Key words	**thousands** (the 4th place from the right), **comparator** (compares two numbers and orders them), **sorting** (putting in order)
Materials	Chalk or tape to draw a sorting network on the floor, number cards (4-digit numbers), place-value chart
Time	5 periods of 40 minutes
Link	Teach alongside chapter 4 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *thousands*, *comparator*, *sorting*. Work through the example *Sort 5, 2, 4 with the network* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The human sorting network	Run the activity below. Move between groups and note who can explain the idea.	Walk through a sorting network, compare 4-digit numbers at each comparator, and check the order.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Turn your class into a sorting network for 3 friends of different heights. Model the first card or step, then let pairs finish.	Try it with three friends. Do they always end up shortest to tallest?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The human sorting network

1. Draw three lanes on the floor with three comparator boxes: lanes 1-2, then 2-3, then 1-2 again.
2. Three children each take a 4-digit number card and stand at the start of a lane.
3. At each box, the two children compare numbers: the smaller goes to the upper lane and the bigger to the lower lane.
4. At the end, check that the numbers are in order. Try again with new cards, then with heights instead of numbers.

Tip: Compare big numbers digit by digit from the left: thousands first, then hundreds. Say it aloud at each box.

Common mistakes to watch for

- **Comparing digits from the right.** Start with the thousands place.
- **Swapping the rule**, so the bigger goes up. Mark 'smaller' on the upper lane.
- **Missing the last comparator.** Every comparison in the network is needed.

Different paces

Support: Use 2-digit numbers first.
Stretch: Can you find three numbers that would come out in the wrong order if the last comparator were removed? (Yes: try 3, 2, 1.)

Exit check and what to note

Two quick questions at the end of the chapter:

1. Which is bigger: 4,072 or 4,207? (Answer: 4,207)

2. At a comparator, 5,310 and 5,103 meet. Which goes to the upper wire? (Answer: 5,103 (the smaller))

Note in the journal: Beginning (needs help to read numbers in the thousands); **Developing** (can read numbers in the thousands with prompts or a partner); **Secure** (can read numbers in the thousands and order big numbers alone); **Extending** (can use a sorting network and explain why).

Answer key

1. 4, 0, 7, 2
2. 3,041; 3,140; 3,401; 3,410
3. b) 3
4. 4, 6, 9
5. 8,510
Build it: Yes: these three comparisons always sort three people
Challenge: No: 3, 2, 1 becomes 2, 3, 1 and then 2, 1, 3, which is not sorted, so the third comparator is needed

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5

TEACHER NOTES · CHAPTER 5 · LEVEL B3

Sharing and Measuring

Coder skill: Fractions and checking a condition

Goes with NCERT maths, Class 4, chapter 5



At a glance	
Students will	name fractions, compare halves and quarters and check a condition before deciding.
Key words	**fraction** (equal parts of a whole), **quarter** (one of 4 equal parts), **condition** (something we check before acting)
Materials	Paper circles (rotis) to fold, counters for sweets, fraction strips
Time	5 periods of 40 minutes
Link	Teach alongside chapter 5 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *fraction*, *quarter*, *condition*. Work through the example *One quarter of 12 laddoos* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Fair or not fair?	Run the activity below. Move between groups and note who can explain the idea.	Fold and share to make halves and quarters, and use an if-else check to decide whether a share is fair.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a checking program that decides if a share is fair. Model the first card or step, then let pairs finish.	Test it: 10 sweets shared as 4, 3, 3. What does the program say?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Fair or not fair?

1. Groups fold paper rotis into halves and quarters and label each part.
2. Share 12 counters into 4 equal groups: each group is one quarter.
3. Act out shares and check the condition: *if all the shares are equal, then 'Fair', else 'Not fair'*.
4. Groups invent a share for another group to check, such as 10 sweets as 4, 3, 3.

Tip: Insist that 'quarter' means 4 *equal* parts. Show a circle cut into 4 unequal parts and ask if each is a quarter.

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Common mistakes to watch for

- **Unequal parts called halves or quarters.** Parts must be equal.
- **Thinking 1/4 is bigger than 1/2** because 4 is bigger than 2. Fold and compare.
- **Checking only one share** instead of all of them before saying 'Fair'.

Different paces

Support: Use halves only, with counters.
Stretch: Share 7 rotis among 4 people fairly. How much does each get? (1 and 3 quarters)

Exit check and what to note

Two quick questions at the end of the chapter:

1. What is one quarter of 20? (Answer: 5)

2. 12 sweets are shared as 4, 4, 4. Is it fair? (Answer: yes, all the shares are equal)

Note in the journal: Beginning (needs help to name fractions); **Developing** (can name fractions with prompts or a partner); **Secure** (can name fractions and compare halves and quarters alone); **Extending** (can check a condition before deciding and explain why).

Answer key

1. b) 3/8
2. a) one half
3. b) no: 15 is odd
4. c) Rs 6
5. 4
Build it: Not fair! (4, 3, 3 are not all the same)
Challenge: 15 (one quarter is 5, so three quarters is 15)

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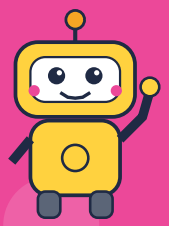
6

TEACHER NOTES · CHAPTER 6 · LEVEL B4

Measuring Length

Coder skill: Shortest paths and loops with turns

Goes with NCERT maths, Class 4, chapter 6



At a glance	
Students will	change metres to centimetres, find a shortest path and draw shapes with a loop and turns.
Key words	**centimetre** (100 make 1 metre), **turn** (change the way Robo faces), **shortest path** (the route with fewest steps)
Materials	Floor grid mat, metre stick and ruler, turn cards (left, right) and forward cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 6 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *centimetre*, *turn*, *shortest path*. Work through the example *Drawing a square with a loop* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Turning Robo	Run the activity below. Move between groups and note who can explain the idea.	Walk programs with turns on the grid, write a program for a rectangle, and measure lengths in m and cm.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a program that draws a rectangle 3 squares long and 2 squares tall. Model the first card or step, then let pairs finish.	Walk it on the floor grid or draw it on squared paper. Does Robo end where it started?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Turning Robo

1. A child is Robo and faces the top of the floor grid. Show the new cards: *turn left*, *turn right*, *forward 2*.
2. Give the program *repeat 4 times: forward 2, turn right*. Robo walks it while the class reads the cards.
3. Ask: 'What shape did Robo walk? Where is Robo facing now?'
4. Pairs write a program for a rectangle and test it on the mat. Then measure a real path in metres and centimetres.

Tip: Turns are from Robo's point of view. Stand behind Robo and face the same way to check left and right.

Common mistakes to watch for

- **Turning the wrong way** because they face Robo. Stand facing the way Robo faces.
- **Moving while turning.** A turn changes the direction only; Robo stays on the same square.
- **Mixing up m and cm.** 1 m = 100 cm, so 2 m 30 cm = 230 cm.

Different paces

Support: Use only forward and turn right on a 3 by 3 grid.
Stretch: Write the shortest program that draws a 3 by 2 rectangle, and one that draws a 2 by 2 square twice.

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many centimetres are in 3 m 40 cm? (Answer: 340)

2. What shape does 'repeat 4 times: forward 3, turn right' draw? (Answer: a square)

Note in the journal: Beginning (needs help to change metres to centimetres); **Developing** (can change metres to centimetres with prompts or a partner); **Secure** (can change metres to centimetres and find a shortest path alone); **Extending** (can draw shapes with a loop and turns and explain why).

Answer key

1. 200 cm
2. b) square
3. 6 moves (right 4, up 2)
4. a) 2 m (200 cm)
5. b) down
Build it: Repeat 2 times: forward 3, turn right, forward 2, turn right; yes, it ends at the start
Challenge: No: it draws 3 sides of a square; Robo needs a 4th side to get back to the start

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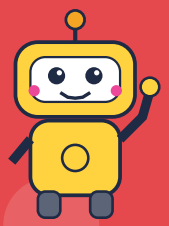
7

TEACHER NOTES · CHAPTER 7 · LEVEL B3

The Cleanest Village

Coder skill: Tracing a machine step by step

Goes with NCERT maths, Class 4, chapter 7



At a glance	
Students will	describe a state, trace a machine press by press and design a simple machine.
Key words	**state** (how things are right now), **toggle** (switch ON to OFF or OFF to ON), **trace** (follow step by step)
Materials	Two paper lamps (ON on one side, OFF on the other), button cards with rules, a trace table on the board
Time	5 periods of 40 minutes
Link	Teach alongside chapter 7 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *state*, *toggle*, *trace*. Work through the example *Two lamps and two buttons* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The lamp machine	Run the activity below. Move between groups and note who can explain the idea.	Act out a two-lamp machine, record the state after each press, and design a machine for others to trace.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Design your own machine with two lamps and two buttons. Decide what each button does. Model the first card or step, then let pairs finish.	Give a friend three presses to trace. Check their final state.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The lamp machine

1. Two children hold paper lamps, both showing OFF. Two more hold buttons: button A *toggles lamp 1*; button B *toggles both lamps*.
2. Call out presses: A, B, B, A. After each press, the lamps change and the class writes the state in a trace table.
3. Before the last press, ask the class to predict the final state.
4. Groups design their own two-button machine and give another group three presses to trace.

Tip: Insist on writing the state after every press. Children who skip steps lose track, just as computers would.

Common mistakes to watch for

- **Writing only the final state** and getting lost. Write the state after every press.
- **Changing the wrong lamp.** Read the button's rule each time.
- **Forgetting that toggle flips:** ON becomes OFF and OFF becomes ON.

Different paces

Support: Use one lamp and one toggle button.
Stretch: Find a sequence of presses that turns both lamps ON from both OFF, using the fewest presses.

Exit check and what to note

Two quick questions at the end of the chapter:

1. A lamp is OFF. A button toggles it. What is it after 3 presses? (Answer: ON)

2. Lamps 1 and 2 are both OFF. Button A toggles lamp 1; button B toggles both. Press A then B. What is the state? (Answer: lamp 1 OFF, lamp 2 ON)

Note in the journal: Beginning (needs help to describe a state); **Developing** (can describe a state with prompts or a partner); **Secure** (can describe a state and trace a machine press by press alone); **Extending** (can design a simple machine and explain why).

Answer key

1. b) both OFF
2. a) A ON, B OFF
3. 1 (press 2)
4. 3
5. after 2: ON, ON; after 1: OFF, ON
Build it: Open answer
Challenge: 2 presses: press 2 (both ON), then press 1 (A OFF)

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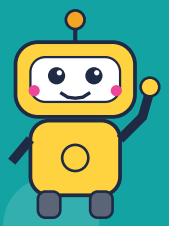
8

TEACHER NOTES · CHAPTER 8 · LEVEL B3

Weigh It, Pour It

Coder skill: If-else decisions and measuring steps

Goes with NCERT maths, Class 4, chapter 8



At a glance	
Students will	compare with a balance, use an if-else rule and plan pouring steps.
Key words	**heavier** (weighs more), **litre** (a unit for liquids), **kilogram** (1,000 grams)
Materials	A simple balance, a 5 L and a 3 L container (or 5-cup and 3-cup jugs) with water or sand, parcel cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 8 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *heavier*, *litre*, *kilogram*. Work through the example *Measure 2 litres with a 5 L jug and a 3 L jug* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The jug puzzle	Run the activity below. Move between groups and note who can explain the idea.	Solve jug puzzles by filling, pouring and emptying, write the steps as an algorithm, and sort parcels with an if-else rule.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write an if-else program for a parcel counter. Model the first card or step, then let pairs finish.	Where do parcels of 1 kg, 3 kg and 2 kg go?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The jug puzzle

1. Show a 5-cup and a 3-cup jug. Challenge: measure exactly 2 cups.
2. Groups try and record each step: fill, pour, empty. (Fill the 5, pour into the 3 until it is full: 2 cups are left in the 5.)
3. Then: measure exactly 4 cups. Groups write their pouring steps as an algorithm.
4. Finish with parcels on the balance: *if the parcel is more than 2 kg, then truck, else bike.*

Tip: Do the jugs outside or over a tray. Using sand or rice instead of water keeps it tidy.

Common mistakes to watch for

- **Pouring without a plan** and losing count. Write each step before doing it.
- **Reading the balance backwards.** The side that goes down is heavier.
- **'More than 2' versus '2 or more':** a 2 kg parcel is not more than 2.

Different paces

Support: Measure 2 cups only, with the teacher modelling the first step.
Stretch: Measure 4 cups with the 5 and 3 jugs in the fewest steps, and compare plans with another group.

Exit check and what to note

Two quick questions at the end of the chapter:

1. With a 5 L and a 3 L jug, how do you get exactly 2 L? (Answer: fill the 5 L jug and pour into the 3 L jug until it is full; 2 L is left)

2. Rule: if more than 2 kg, truck, else bike. Where does a 2 kg parcel go? (Answer: bike)

Note in the journal: Beginning (needs help to compare with a balance); **Developing** (can compare with a balance with prompts or a partner); **Secure** (can compare with a balance and use an if-else rule alone); **Extending** (can plan pouring steps and explain why).

Answer key

1. a) apple
2. b) two hands
3. 1,000
4. A, B, C
5. 2 litres
Build it: 1 kg: bike; 3 kg: truck; 2 kg: bike (2 is not more than 2)
Challenge: Fill 5; pour into 3 (2 left in 5); empty 3; pour the 2 into 3; fill 5; pour into 3 until full (1 goes); 4 litres stay in the 5 L jug

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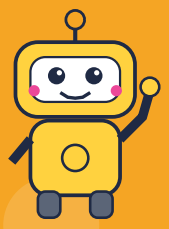
9

TEACHER NOTES · CHAPTER 9 · LEVEL B4

Equal Groups

Coder skill: Loops inside loops

Goes with NCERT maths, Class 4, chapter 9



At a glance	
Students will	count with a nested loop, read an array and write a loop inside a loop.
Key words	**nested loop** (a loop inside a loop), **array** (objects in rows and columns), **row** (a line going across)
Materials	Counters or seeds, squared paper, egg trays or ice-cube trays, nested loop cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 9 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *nested loop*, *array*, *row*. Work through the example *Stars in rows* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Planting the garden	Run the activity below. Move between groups and note who can explain the idea.	Act out a nested loop, count with rows and columns, and write nested loops for real arrays.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Plan a garden with 4 rows and 5 plants in each row. Write it as a nested loop. Model the first card or step, then let pairs finish.	How many seeds do you need?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Planting the garden

1. Show the nested loop: *repeat 3 times: [repeat 4 times: plant a seed], start a new row.*
2. A child 'plants' counters on a grid while the class counts the inner loop (1, 2, 3, 4) and the outer loop (row 1, 2, 3).
3. Ask: 'How many seeds? How many times did the inner loop run altogether?' (3 x 4 = 12)
4. Pairs write a nested loop for an egg tray or a classroom seating plan and check it by counting.

Tip: Use two hands: one counts the rows (outer loop), the other counts within a row (inner loop).

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Common mistakes to watch for

- **Adding instead of multiplying**: 3 rows of 4 is 12, not 7.
- **Running the inner loop only once** in total. It runs fully every time the outer loop repeats.
- **Forgetting 'start a new row'**, so all the seeds end up in one long line.

Different paces

Support: Use 2 rows of 3 with real counters.
Stretch: Write a nested loop that draws a 5 by 5 square of stars, then one that draws only the border.

Exit check and what to note

Two quick questions at the end of the chapter:

1. An outer loop repeats 3 times; inside, an inner loop repeats 5 times. How many times does the inner action happen? (Answer: 15)

2. How many eggs are in a tray with 4 rows of 6? (Answer: 24)

Note in the journal: Beginning (needs help to count with a nested loop); **Developing** (can count with a nested loop with prompts or a partner); **Secure** (can count with a nested loop and read an array alone); **Extending** (can write a loop inside a loop and explain why).

Answer key

1. 15
2. 24
3. a) 3 x 6 = 18
4. 30
5. 2 rows of 3 circles
Build it: Repeat 4 times: [repeat 5 times: plant a seed], start a new row; 20 seeds
Challenge: 6 claps and 3 stamps

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10

TEACHER NOTES · CHAPTER 10 · LEVEL B3

Elephants, Tigers and Leopards

Coder skill: Clue tables and data

Goes with NCERT maths, Class 4, chapter 10



At a glance	
Students will	use a clue table, read a data table and write my own clue puzzle.
Key words	**clue table** (a grid to track clues), **data** (facts we collect and count), **logic** (reasoning step by step)
Materials	Clue-table grids drawn on paper or the board, animal picture cards, a small data table
Time	5 periods of 40 minutes
Link	Teach alongside chapter 10 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *clue table*, *data*, *logic*. Work through the example *Who lives in which enclosure?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Solve it with a clue table	Run the activity below. Move between groups and note who can explain the idea.	Solve a logic puzzle with ticks and crosses, then write and test their own clues.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Make a clue puzzle for a friend: 3 children, 3 favourite fruits. Write two clues that are enough to solve it. Model the first card or step, then let pairs finish.	Can your friend solve it with a clue table? Are both clues needed?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Solve it with a clue table

1. Draw a 3 by 3 clue table: elephant, tiger, leopard against enclosures A, B, C.
2. Read one clue at a time. The class puts crosses where something is impossible.
3. When a row has only one empty box, put a tick there and cross out the rest of its column.
4. Pairs make their own clue puzzle (3 children, 3 fruits) and test whether their clues are enough.

Tip: Ticks and crosses are a record. Ask children to say which clue gave each mark.

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Common mistakes to watch for

- **Putting a tick too early**, before the other boxes in that row are crossed out.
- **Forgetting to cross out the rest of the column** after a tick.
- **Writing clues that are not enough**, so two answers are still possible.

Different paces

Support: Use a 2 by 2 clue table with two clues.
Stretch: Write a puzzle that needs exactly three clues, where removing any clue gives more than one answer.

Exit check and what to note

Two quick questions at the end of the chapter:

1. In a clue table, a row has crosses in two of three boxes. What goes in the third box? (Answer: a tick)

2. Why should you cross out the rest of a column after putting a tick? (Answer: each item can have only one match)

Note in the journal: Beginning (needs help to use a clue table); **Developing** (can use a clue table with prompts or a partner); **Secure** (can use a clue table and read a data table alone); **Extending** (can write my own clue puzzle and explain why).

Answer key

1. Asha blue, Ben red, Chitra green
2. 25
3. b) tiger
4. c) tiger
5. 300 kg
Build it: Open answer
Challenge: Leopard 1, deer 2, tiger 3, elephant 4

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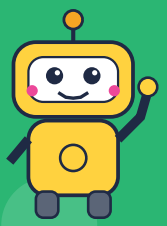
11

TEACHER NOTES · CHAPTER 11 · LEVEL B4

Fun with Symmetry

Coder skill: Mirror patterns and repeat with turns

Goes with NCERT maths, Class 4, chapter 11



At a glance	
Students will	find lines of symmetry, complete a mirror pattern and draw a design with repeat and turn.
Key words	**symmetry** (both halves match), **line of symmetry** (the fold line), **mirror image** (a flipped copy)
Materials	Squared paper, mirrors (if available), paper to fold and cut, rangoli dot grids
Time	5 periods of 40 minutes
Link	Teach alongside chapter 11 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *symmetry*, *line of symmetry*, *mirror image*. Work through the example *Complete the mirror pattern* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Mirror and rangoli	Run the activity below. Move between groups and note who can explain the idea.	Fold and cut symmetrical shapes, complete mirror patterns, and draw designs with repeat and turn.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a loop that draws a flower with 6 petals. Model the first card or step, then let pairs finish.	Draw your flower. Does it have lines of symmetry? Mark them.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Mirror and rangoli

1. Fold a paper in half, cut a shape along the fold, and open it. Ask: 'Where is the line of symmetry?'
2. On squared paper, pairs complete half-drawn patterns across a mirror line, counting squares from the line.
3. Show the loop *repeat 4 times: draw a petal, turn right*. Pairs draw a 4-petal design, then a 6-petal one.
4. Mark the lines of symmetry on each design.

Tip: Count squares *from the mirror line*, not from the edge of the paper. The mirror point is the same distance away.

Common mistakes to watch for

- **Copying instead of mirroring**, so the shape faces the same way on both sides.
- **Counting from the edge** instead of from the mirror line.
- **Turning a different amount each time** in the petal loop, so the design is uneven.

Different paces

Support: Use small shapes of 2 or 3 squares next to the mirror line.
Stretch: Design a pattern with exactly 2 lines of symmetry, and write the loop that draws it.

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many lines of symmetry does a square have? (Answer: 4)

2. A shaded square is 2 squares left of the mirror line. Where is its mirror image? (Answer: 2 squares right of the line, in the same row)

Note in the journal: Beginning (needs help to find lines of symmetry); **Developing** (can find lines of symmetry with prompts or a partner); **Secure** (can find lines of symmetry and complete a mirror pattern alone); **Extending** (can draw a design with repeat and turn and explain why).

Answer key

1. A and H
2. c) F
3. c) 4
4. 4
5. b) left
Build it: Repeat 6 times: draw a petal, turn; the flower has lines of symmetry through the petals
Challenge: B2

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12

TEACHER NOTES · CHAPTER 12 · LEVEL B4

Ticking Clocks and Turning Calendar

Coder skill: Repeat until and cycles

Goes with NCERT maths, Class 4, chapter 12



At a glance	
Students will	use a repeat-until loop, read clocks to the quarter hour and work out days with the week cycle.
Key words	**repeat until** (loop until something is true), **cycle** (comes back to the start), **minute** (60 make an hour)
Materials	Kit clock, sand timer or a phone stopwatch, a calendar page, repeat-until cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 12 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *repeat until*, *cycle*, *minute*. Work through the example *How many 15-minute steps from 3:00 to 4:00?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Repeat until the timer stops	Run the activity below. Move between groups and note who can explain the idea.	Use repeat-until loops with a timer, a clock and a calendar, and say when the loop stops.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a 'repeat until' program for brushing your teeth. Model the first card or step, then let pairs finish.	Why is 'repeat until' better here than 'repeat 12 times'?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Repeat until the timer stops

1. Start a 1-minute timer. The class does *repeat until the timer stops: clap twice, tap knees*.
2. Ask: 'How many times did we repeat? Did we know before we started?' (No: we stopped when the condition came true.)
3. Use the clock: move the minute hand in 15-minute steps from 3:00 *until* it is 4:00. Count the steps.
4. On the calendar, count on in weeks: 'repeat until we reach the 30th: add 7 days'.

Tip: Contrast with 'repeat 10 times'. Ask which suits brushing teeth, and why.

Common mistakes to watch for

- **Thinking repeat until must know the number of repeats.** It stops when the condition becomes true.
- **Reading the hour hand wrongly** at quarter past and quarter to.
- **Forgetting the cycle**: after 12 comes 1, and after Sunday comes Monday.

Different paces

Support: Count 15-minute steps on the clock face with a pointer.
Stretch: Start at 9:45 and add 15 minutes until it is 11:00. How many steps? (5)

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many 15-minute steps are there from 2:00 to 3:30? (Answer: 6)

2. Today is Friday. What day is it 7 days from now? (Answer: Friday)

Note in the journal: Beginning (needs help to use a repeat-until loop); **Developing** (can use a repeat-until loop with prompts or a partner); **Secure** (can use a repeat-until loop and read clocks to the quarter hour alone); **Extending** (can work out days with the week cycle and explain why).

Answer key

1. 3
2. A 4:15, B 7:45, C 10:30
3. 150
4. c) Wednesday
5. 5
Build it: We care about the time, not the number of repeats; the loop stops when 2 minutes are over
Challenge: Friday (26 days later = 3 full weeks + 5 days)

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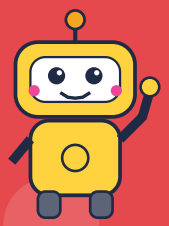
13

TEACHER NOTES · CHAPTER 13 · LEVEL B3

The Transport Museum

Coder skill: Decision trees: sorting by questions

Goes with NCERT maths, Class 4, chapter 13



At a glance	
Students will	follow a decision tree, ask good yes/no questions and build my own tree.
Key words	**decision tree** (yes/no questions that sort), **branch** (a path in the tree), **classify** (put into the right group)
Materials	Vehicle and animal picture cards, chart paper, question cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 13 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *decision tree*, *branch*, *classify*. Work through the example *Sorting the museum's vehicles* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Twenty questions tree	Run the activity below. Move between groups and note who can explain the idea.	Sort picture cards with yes/no questions, draw the decision tree, and build their own tree.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Design a decision tree two levels deep (two questions on every path) that sorts a fish, a parrot, a cow and a frog. Model the first card or step, then let pairs finish.	Test your tree with all four animals. Does each end up alone?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Twenty questions tree

1. Spread vehicle cards on a table. Ask one yes/no question, such as 'Does it run on rails?', and split the cards into two piles.
2. Ask a new question for each pile until every card is alone. Draw the tree on chart paper as you go.
3. Ask: 'Which question split the cards best? Which was not useful?'
4. Pairs build a two-level tree for fish, parrot, cow and frog, and test it with all four cards.

Tip: A good question splits a pile roughly in half. 'Is it red?' is rarely useful for sorting vehicles.

Common mistakes to watch for

- **Questions that are not yes/no**, such as 'What colour is it?'.
- **Questions that do not split the pile**, where every card gives the same answer.
- **Leaving two items together** at the end of a branch. Each item must end up alone.

Different paces

Support: Sort 4 cards with two questions given by the teacher.
Stretch: Build a tree for 8 cards that uses only 3 levels of questions.

Exit check and what to note

Two quick questions at the end of the chapter:

1. In a decision tree, what kind of questions do we ask? (Answer: yes/no questions)

2. Give one question that separates a parrot from a cow. (Answer: for example 'Can it fly?' or 'Does it have feathers?')

Note in the journal: Beginning (needs help to follow a decision tree); **Developing** (can follow a decision tree with prompts or a partner); **Secure** (can follow a decision tree and ask good yes/no questions alone); **Extending** (can build my own tree and explain why).

Answer key

1. b) Train
2. b) Does it have an engine?
3. 16
4. c) boat (it does not run on roads)
5. Engine: auto-rickshaw, scooter. No engine: cycle, bullock cart
Build it: Example: Does it live in water? Yes: Does it live only in water? (yes fish, no frog). No: Can it fly? (yes parrot, no cow)
Challenge: 3 (8, then 4, then 2, then 1)

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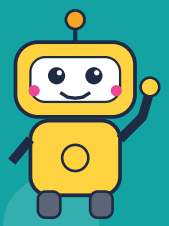
14

TEACHER NOTES · CHAPTER 14 · LEVEL B2 review

Data Handling

Coder skill: Collecting and showing data

Goes with NCERT maths, Class 4, chapter 14



At a glance	
Students will	make tally marks, read a bar graph and answer questions with data.
Key words	**tally marks** (lines for quick counting), **bar graph** (bars that show counts), **survey** (asking a question to many people)
Materials	Tally chart on the board, squared paper for bar graphs, coloured pencils
Time	5 periods of 40 minutes
Link	Teach alongside chapter 14 of the NCERT Class 4 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *tally marks*, *bar graph*, *survey*. Work through the example *Our class's favourite fruit* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Class survey	Run the activity below. Move between groups and note who can explain the idea.	Collect votes with tally marks, draw a bar graph, and write what the data shows.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Plan a class survey. Put the steps in order. Model the first card or step, then let pairs finish.	Carry out your survey and write one thing your graph shows.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Class survey

1. Choose a question with the class, such as 'Which fruit do you like most?', and list 4 choices.
2. Each child votes once while a recorder makes tally marks, crossing every fifth mark.
3. Groups count the tallies and draw a bar graph on squared paper with a title and labels.
4. Each group writes two sentences about what the graph shows and one question the data cannot answer.

Tip: Ask 'How do we know everyone voted once?' Checking that the total equals the class size is a simple data check.

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Common mistakes to watch for

- **Tally marks without the crossing fifth mark**, which makes counting slow.
- **Bars of unequal width** or no labels, which make the graph hard to read.
- **Claiming more than the data shows**, such as 'everyone hates mangoes' when 3 children chose them.

Different paces

Support: Use 3 choices and a ready-made graph frame.
Stretch: Compare your class's graph with another class's. Write one similarity and one difference.

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many does this tally show: one bundle of five and three more marks? (Answer: 8)

2. In a bar graph, how do you find the most popular choice? (Answer: it has the tallest bar)

Note in the journal: Beginning (needs help to make tally marks); **Developing** (can make tally marks with prompts or a partner); **Secure** (can make tally marks and read a bar graph alone); **Extending** (can answer questions with data and explain why).

Answer key

1. 7
2. c) banana
3. 26
4. 3
5. Two groups of five (each four lines with one across) and two more lines
Build it: Ask a question; make tally marks; count the tallies; draw a bar graph
Challenge: Apple (banana now has 8, apple has 6)

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Observation journal

Class 4 ____ Section ____ Term ____ Chapters _____. Circle one level per skill: **B**eginning, **D**eveloping, **S**ecure, **E**xtending. You may photocopy this page.

Student	Following steps	Patterns	Breaking down	Logic and rules	Notes
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	

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Term summary for the report card

One copy per child each term. Fill it from the observation journal and the child's self-checks. You may photocopy this page.

Name _____ Class _____ Section _____ Term _____

Area	Level (circle one)	Teacher comment
Computational thinking (overall)	B D S E	
Tracing and writing programs	B D S E	
Patterns and checking	B D S E	
Logic, rules and decisions	B D S E	
Coding on paper ('build it' tasks)	B D S E	

Writing comments

Write one strength and one next step, in words a parent understands. For example: *Explains steps clearly and fixes bugs in routes; next step: spotting the core of a pattern.*

Class teacher _____ Date _____

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